Impact of Advanced AI Agents on the Enforcement-Circumvention Balance in Export Controls

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Abstract

U.S. export controls on advanced AI chips are failing to prevent large-scale diversion to restricted destinations, and the gap between what regulators can detect and what they can actually stop is growing. Recent prosecutions describe diversion networks that exported or attempted to export at least $160 million in NVIDIA H100/H200 GPUs to restricted destinations within a few months (Department of Justice, 2025), while the Bureau of Industry and Security (BIS) remains resource constrained (Government Accountability Office, 2025). At the same time, autonomous AI agents are rapidly improving at multistep work such as document review, cross-jurisdictional entity matching, and planning and execution, all capabilities relevant to both enforcement and circumvention workflows (Kwa et al., 2025). We focus on diversion of controlled AI chips and the enforcement task chain around licensing, outbound checks, and end use monitoring, and we analyze how the enforcement–circumvention balance may evolve as AI agents are adopted on both sides. This paper makes three contributions. First, we distinguish the constraints AI can relax (information, triage, and coordination) from constraints that remain binding (legal authority, institutional throughput, and control at physical and financial chokepoints) on both sides of the enforcement–circumvention contest. Second, we assess domains of agentic capability and map their likely effects onto enforcement and circumvention task chains using benchmark evidence and enforcement-action records. Third, we lay out four plausible 2027–2029 scenarios based on how quickly governments and smuggling networks adopt agents, and we list real-world signals that would tell us which path we are on. We find that while capability gains are broadly symmetric, the default dynamics favor circumvention due to three structural asymmetries. Adversaries can adopt AI agents more quickly than the U.S. government, because government procurement and deployment is slow. Adversaries can also tolerate higher failure rates than the U.S. government and exporters. And AI-enhanced detection of export violations is likely to scale faster than institutional capacity to convert leads into enforcement actions (e.g., arrests and penalties), because enforcement is fragmented across agencies and jurisdictions, slowing handoffs and escalation. Without reforms that tighten the link between detection and action at key chokepoints (e.g., exporters/distributors, logistics, and payments), expanded AI tooling may increase enforcement workload without proportionally reducing diversion.

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1. Introduction

1.1. Problem

Ananya *example comment* U.S. export controls on advanced AI chips constitute a central instrument of national security strategy, intended to slow adversary access to frontier AI capabilities. The Bureau of Industry and Security (BIS) administers the Export Administration Regulations (EAR), which govern the export, reexport, and in-country transfer of dual-use technologies, including advanced semiconductors and AI accelerators (Bureau of Industry and Security, 2022; Heim, 2025). These controls rely on licensing requirements, end-use restrictions, and post-shipment compliance mechanisms to constrain access to high-performance compute, increasingly viewed as a prerequisite for national technological advantage (Kendler et al., 2026).

The current balance already favors circumvention. BIS operated in fiscal year 2023 with an annual budget of approximately $191 million (with a requested increase to $303 million for FY2026) and employed 585 staff, of whom 52 positions were vacant, a roughly 9 percent vacancy rate (Government Accountability Office, 2025). The Government

Accountability Office has found that BIS lacks a "comprehensive long-term workforce plan amid significant growth in licenses and end-use checks" (Government Accountability Office, 2025). On the circumvention side, Operation Gatekeeper, a Department of Justice enforcement action disclosed in December 2025, revealed a network that trafficked over $160 million worth of NVIDIA H100 and H200 GPUs to China over just seven months using straw-buyer methodology, front companies, physical relabeling, and wire transfers routed through intermediate jurisdictions (Department of Justice, 2025). Industry reporting indicates that at least eight similar Chinese procurement networks are operating, each handling volumes exceeding $100 million (Allen, 2025; Liu, 2024).

A further factor complicates this asymmetry. Autonomous AI agents, software systems that plan and execute multi-step actions without continuous human oversight, have improved substantially on standardized benchmarks over the past 18 months. SWE-bench Verified, which measures autonomous software engineering, improved from approximately 12 percent to 79 percent between 2024 and late 2025 (Scale AI, 2026). GAIA, measuring general task completion in simulated environments, rose from 15 percent to 74.5 percent (Mialon et al., 2024). OSWorld desktop task completion improved from under 20 percent to 76.3 percent, the first result exceeding human performance (Simular, 2026). Task completion horizons (as defined by Kwa et al., the maximum duration of tasks an agent can complete with greater than 50 percent reliability), the duration of tasks agents can reliably execute, are doubling approximately every seven months (Kwa et al., 2025). These capabilities in document processing, entity resolution, code generation, and multi-step planning map directly onto both enforcement and circumvention workflows.

1.2. Gaps in Literature

The offense-defense literature in international security (Jervis 1978; Glaser and Kaufmann 1998; Van Evera 1998) and its adaptation to cybersecurity (Slayton 2017; Garfinkel and Dafoe 2019) provide the structural framework for this analysis. Recent work on AI and national security identifies the risk that AI systems may eventually act as autonomous operatives in procurement and logistics. No existing work systematically maps current agentic AI capabilities onto the enforcement and circumvention task chains documented in BIS enforcement actions and DOJ prosecutions. The cybersecurity analogy shapes our approach but requires careful adaptation, because the structural dynamics differ in ways that matter for the analysis. In cybersecurity, a successful exploit can be repeated until the vulnerability is patched, while in export controls, a successful diversion permanently transfers physical goods, but the network that executed it accumulates exposure with each transaction and can be disrupted through prosecution. Conversely, patching a software vulnerability protects all systems simultaneously, whereas interdicting one smuggling network does nothing to prevent the next. The variables that matter in export controls are therefore reconstitution speed, how quickly a disrupted network can rebuild, and detection latency, how many successful diversions occur before a network is identified, rather than vulnerability surface area as conventionally understood in cybersecurity. Two additional literatures inform our framework. Farrell and Newman's (2019) "weaponized interdependence" framework offers a competing structural explanation for the dynamics we analyze. Since export controls are fundamentally chokepoint strategies, exploiting network centrality in semiconductor supply chains, any analysis applying offense-defense theory to this domain should consider whether the chokepoint structure of global supply chains creates enforcement advantages that the offense-defense framing underweights. The sanctions effectiveness literature, particularly Itskhoki and Ribakova (2024), offers a different evaluative lens altogether, one that asks not whether enforcement achieves perfect control but whether it keeps diversion costly, risky, and limited in scale. We engage this framing explicitly in Section 5.

1.3. Contributions

This paper makes three contributions. First, we distinguish informational constraints that AI can relax (screening, triage, documentation) from structural constraints that remain binding regardless of computational capability (physical verification, legal authority, jurisdictional reach), and show that the constraints AI relaxes on the circumvention side are closer to current operational limits than those it relaxes on the enforcement side. Second, we assess seven agentic AI capability domains and map their differential impact onto enforcement and circumvention task chains using benchmark evidence and enforcement-action records. Third, we develop four 2027-2029 scenarios structured around adoption postures, and identify three observable indicators, diversion success rate, detect-to-hold conversion speed, and the share of diversion shifting to post-shipment channels, that would signal which trajectory is materializing.

1.4. Scope and Definitions

By "agentic AI," we mean AI systems capable of autonomously planning and executing multi-step actions in pursuit of a goal, without continuous human oversight. Unlike conventional AI tools that perform discrete tasks upon instruction, agentic systems can decompose complex objectives into subtasks, execute them sequentially across digital environments, and adapt their approach based on intermediate results. A concrete illustration: an agentic system could, in principle, coordinate shipping logistics across jurisdictions, generate internally consistent corporate documentation for a shell company, and monitor regulatory changes to trigger network reorganization, without a human operator directing each step.

Beyond serving as tools that amplify human operators' productivity, advanced AI agents introduce what Withers (2025) and Rasser (2026) term "functional risk," the possibility that AI systems themselves act as autonomous operatives capable of executing cyber operations, managing supply chains, or conducting procurement with minimal human oversight. While no documented case of fully autonomous AI-managed smuggling exists as of this writing, the trajectory of agentic capabilities suggests this risk warrants serious analytical attention. We treat functional risk as a plausible near-term development rather than an established fact, and flag it accordingly in our assumption ledger (Section 5.2.4).

We adopt the terminology of offense-defense theory (Garfinkel & Dafoe, 2019). Circumvention actors occupy the offensive role, seeking to penetrate or bypass controls, while regulators, enforcement agencies, and compliance systems occupy the defensive role. This framing leverages the core insight that structural conditions can systematically favor one side, and that technological change can shift the balance.

1.5. Methodology

Our analysis draws on DOJ prosecution materials, BIS enforcement actions, GAO audits, published AI capability benchmarks (SWE-bench Verified, GAIA, METR task-horizon data), and semi-structured expert elicitations with former U.S. export control officials, industry compliance professionals, and researchers working on adjacent problems in AI governance and semiconductor policy. The confidence scores in Table 3, bottleneck composition estimates in Figure 1, and scenario net-advantage scores in Table 7 were produced through independent scoring by each co-author followed by consensus discussion, informed by both documentary evidence and expert input. We report these as structured qualitative judgments rather than empirically calibrated measurements, and specify disconfirmation criteria in Section 7.8. Interviewees included former BIS officials with direct enforcement experience, compliance professionals at major semiconductor distributors, and researchers working on adjacent problems in AI governance and semiconductor policy. We do not name individual interviewees but characterize their institutional positions where relevant to the weight of specific claims.

[1]The term "binding" appears in three related senses: **structurally binding constraint** is the theoretical concept within the bottleneck persistence framework; **binding layers** is the operational manifestation in the enforcement pipeline; and **binding actions** are the specific enforcement outputs. In each case, "binding" denotes a step or condition that materially constrains diversion rather than merely detecting or flagging it.

2. The Enforcement Architecture: Structure and Constraints

Export control enforcement can fail in two distinct ways. It can fail to identify suspicious transactions because data is fragmented, entities are difficult to resolve across jurisdictions, or screening capacity is thin. Or it can identify suspicious transactions and still fail to stop them, because physical verification is scarce, jurisdictional authority weakens after the first export hop, and institutional handoffs are slow. These are different failure modes, and they respond to different interventions. AI is likely to address the first more than the second. This section establishes that distinction by mapping the current enforcement architecture and identifying where its binding constraints actually sit. Sections 3 through 5 then use that map to assess how AI adoption changes the balance. (Government Accountability Office, 2025; Center for Strategic and International Studies, 2022).

2.1. Government Enforcement Architecture

U.S. export enforcement is centered in the Bureau of Industry and Security (BIS), but it operates through linked functions. BIS administers and interprets the Export Administration Regulations (EAR) through its rulemaking and licensing architecture, while Export Enforcement, primarily through the Office of Export Enforcement (OEE), investigates suspected violations of the EAR, including diversion networks, falsified documentation, and illicit procurement.

According to practitioner interviews conducted for this study, OEE maintains roughly 150 sworn criminal investigators, largely based in domestic field offices (Practitioner interviews, 2025–2026). A smaller overseas cadre of Export Control Officers (ECOs), roughly 9–11, is posted at U.S. embassies to support end-use checks and local engagement (Practitioner interviews, 2025–2026).

BIS uses two primary verification tools. Pre-License Checks (PLCs) assess whether a proposed end user and end use appear credible before a license is approved. Post-Shipment Verifications (PSVs) attempt to confirm that exported items are where they are supposed to be and used as represented. Practitioner interviews suggest PLC volumes on the order of $\sim$100 per year, with most end-use checks returning formally favorable results, though a meaningful minority involve non-cooperation, misunderstandings, or suspected diversion (Practitioner interviews, 2025–2026).

Export shipments (valued above \$2,500 per Schedule B item or requiring an export license) are also documented through Electronic Export Information (EEI) filings submitted via the Automated Export System (AES) (Code of Federal Regulations, 2024). As a result, many U.S. exports of controlled AI chips will appear in AES records and are used for outbound targeting and pattern detection, but its usefulness depends on exporter-reported accuracy. AES primarily captures the initial U.S. export transaction. Subsequent resales or redistribution by intermediaries outside the U.S. falls outside the system (Code of Federal Regulations, 2024; Practitioner interviews, 2025–2026).

Operationally, enforcement hinges on several chokepoints. The first is the original manufacturer or designer. The second is the distribution layer: authorized distributors, resellers, integrators, and logistics intermediaries where hardware moves after the initial sale (Practitioner interviews, 2025–2026). The third is the downstream compute layer, including cloud and Infrastructure-as-a-Service providers, where chips become difficult to trace through trade documentation alone (Heim, 2025).

The ease of control varies by item type. Semiconductor manufacturing equipment is relatively controllable because it comes from a small supplier base and often requires installation and servicing that generate visibility. Commodity chips are harder to control because they are low-value, high-volume, and easy to conceal within legitimate trade flows. Advanced AI accelerators fall in between: compact and extremely high value, difficult to identify reliably at borders, and increasingly opaque once absorbed into reseller chains or cloud provisioning ecosystems (Practitioner interviews, 2025–2026).

2.2. Exporter Compliance Architecture

Export controls rely heavily on private-sector compliance. Under the EAR, exporters are responsible for classifying items, screening counterparties against restricted-party lists, identifying prohibited end uses and destinations, and escalating transactions when diversion-related red flags cannot be resolved. BIS "Know Your Customer" guidance sets expectations that exporters should investigate warning signs even when a transaction does not automatically require a license.

In practice, large upstream firms (chip designers, server manufacturers, major distributors) often operate structured compliance programs that include customer verification, end-user certificates, restricted-party screening, internal red-flag reviews, and license applications where required. Many also use third-party intelligence and compliance providers (e.g., Sayari, Altana, and similar vendors) for entity resolution, beneficial ownership analysis, sanctions screening, and trade-linked risk signals (Practitioner interviews, 2025–2026). These tools can materially improve screening and investigations, but their outputs remain probabilistic and depend on the availability, quality, and freshness of underlying records.

Two features shape exporter-side compliance in practice. First, compliance expectations are often articulated through guidance and "reasonable" due diligence rather than prescriptive operational standards, leaving firms discretion in how they operationalize screening and escalation. Second, as products move downstream to regional distributors, resellers, and smaller intermediaries, diligence and data access often become less consistent. As a result, many compliance programs are optimized to manage legal liability and commercial risk rather than to detect sophisticated diversion networks. Programs that are sufficient for defensibility in routine transactions may still be ill-suited to identifying coordinated front-company procurement, falsified end-user claims, or multi-hop reexport pathways.

The policy environment has also shifted in ways that matter for enforcement. In May 2025, the Department of Commerce rescinded the Biden Administration's AI Diffusion Rule before its compliance date and indicated that a replacement approach would follow (Kirkland & Ellis LLP, 2025; Wolber et al., 2025). Since then, the policy direction has moved away from centralized quota-style allocation and toward a mix of negotiated bilateral arrangements, compliance expectations for firms, and broader chip-control enforcement measures (Kendler et al., 2026). For this paper, the main implication is simple: enforcement depends increasingly on monitoring, verification, and follow-through across distributed commercial networks rather than on a single centralized allocation system. That makes the practical enforcement architecture described above even more important.

2.3. Shared Bottlenecks Across the System

Viewed as a system, government enforcement plus delegated private compliance, export control effectiveness is constrained by bottlenecks that fall into three broad categories.

A. Information and linkage bottlenecks. Fragmented data and high triage costs. Enforcement and compliance teams operate across siloed datasets. Center for Strategic and International Studies (2022) found that officers spend roughly 80 percent of their time locating and reconciling data rather than analyzing it. AI can compress this overhead, but the gain is in triage speed, not in the quality or completeness of the underlying records.

Classification and misreporting risk. Visibility in systems like AES depends on exporter-reported descriptions, values, and classifications. For fast-evolving advanced hardware and complex configurations, misclassification, accidental or deliberate, can be difficult to detect at scale, limiting the reliability of structured records.

Probabilistic identity resolution. Vendor and internal tools can map ownership networks and flag risk signals, but newly formed shell entities, nominee structures, and opaque jurisdictions can remain unresolved until they leave stronger traces in registries or trade data.

B. Physical and traceability bottlenecks. Inspection and targeting throughput limits. Outbound enforcement is a needle-in-a-haystack problem. CBP screens millions of export shipments annually; BIS maintains roughly 150 domestic investigators and 9–11 overseas officers. AI can sharpen the needle, improving which shipments get flagged, but it cannot expand the haystack's physical inspection capacity.

Downstream visibility decay. Monitoring follows a visibility gradient: strong at the initial U.S. export, weak at the first foreign resale, and often nonexistent by the second or third transfer. Coordinated diversion exploits precisely this gradient, concentrating activity in the low-visibility segments of the supply chain (Heim, 2025; Practitioner interviews, 2025–2026).

Weak chain-of-custody and end-to-end traceability. Even when serial numbers exist, end-to-end tracking across resale, integration into servers, and deployment in data centers is often incomplete. This sustains a detect→locate gap: suspicion can rise faster than verifiable location can be established (Heim, 2025; Practitioner interviews, 2025–2026).

C. Institutional and legal bottlenecks. Burden-of-proof asymmetry and reconstitution speed. Circumvention actors can create new entities quickly and cheaply, while formal enforcement actions require evidence development, coordination, and procedural steps. This creates a structural speed disadvantage (Practitioner interviews, 2025–2026).

Jurisdictional barriers to evidence. Overseas end-use checks are often voluntary, and in some jurisdictions—including China—investigators may lose access to records once goods enter legal environments that restrict data sharing or investigative cooperation. In these cases, the bottleneck is legal access to evidence, not analytic capacity.

Interagency seams and adoption lag. Enforcement often requires handoffs across licensing, customs, investigations, and prosecution. Coordination cycles can be slow, and deploying new tools inside government environments is constrained by security and compliance processes that typically move more slowly than commercial adoption (Government Accountability Office, 2025; Center for Strategic and International Studies, 2022).

2.4. Implications for AI-Assisted Enforcement

This architecture suggests that AI will most reliably improve the front end of enforcement and compliance: triage, document processing, entity resolution, and anomaly detection because these are primarily information-processing tasks. But many of the constraints that determine whether enforcement can rapidly disrupt diversion networks lie elsewhere: downstream visibility decay, limited physical inspection and traceability, jurisdictional barriers to evidence collection, and procedural and coordination lags (Government Accountability Office, 2025; Center for Strategic and International Studies, 2022; Practitioner interviews, 2025–2026). The key question for the sections that follow is whether AI can relax the structural bottlenecks, downstream visibility, physical verification, jurisdictional barriers, coordination lags, that determine whether detection leads to interdiction. The capability assessments in Section 4 and the scenario analysis in Section 5 are organized around this question.

3. Circumvention Pathways, Scale, and Operational Bottlenecks

Unlike resource-constrained enforcement agencies, circumvention networks benefit from access to state-directed funding and strong financial incentives created by black-market price premiums. This section maps the dominant diversion pathways, assesses how agentic AI affects each, and identifies the operational bottlenecks that currently constrain circumvention, distinguishing those that AI can relax from those that remain structurally binding.

3.1. Dominant Diversion Pathways

Straw-buyer networks with domestic warehouses. As documented in Operation Gatekeeper, smuggling networks utilize U.S.-registered entities posing as legitimate domestic end users to procure chips from U.S. distributors. Chips are aggregated in domestic warehouses where workers physically remove NVIDIA serial number labels and branding, then relabel as fictitious brands and describe them as "generic computer parts" (Department of Justice, 2025). This physical alteration significantly complicates tracking of origin and destination.

Phantom data centers in transshipment hubs. Shell companies in transshipment hub jurisdictions such as Malaysia, Thailand, Vietnam, and the UAE purchase full server racks ostensibly for local cloud provision. Facilities are built to specification to satisfy initial inspections. Following inspector departure, centers are liquidated and racks disassembled. The suitcase method for Blackwell-series GPUs involves disassembling server chassis into individual GPU modules worth tens of thousands of dollars each, then transporting them via land or short-haul flights into southern China, bypassing maritime cargo screenings. Singapore accounts for 18 to 22 percent of NVIDIA revenue despite representing under 1 percent of actual end-use demand, a discrepancy that signals significant transshipment activity (CNBC, 2025).

Shell company proliferation. The scale of shell company use in sanctions and export control evasion is well documented across contexts. Bilousova et al. (2024) found that approximately $4 billion flowed to Russia through over 6,000 shell companies, many featuring minimal web presence, virtual office addresses, reliance on mobile phones, and inability to demonstrate real operations. Similar structures characterize Chinese semiconductor procurement networks. These entities can be established rapidly and cheaply, and their proliferation creates a combinatorial challenge for enforcement agencies.

Black-market economics. Market prices sustain and incentivize circumvention. B200 rack prices in China are reported at $420,000 to $490,000 versus $280,000 to $330,000 in the United States, representing an approximately 50 percent black-market premium and profit margins exceeding $100,000 per transaction (Financial Times, 2025; DIGITIMES, 2025). These margins sustain continued investment in circumvention infrastructure, including the development and deployment of AI-assisted operational tools.

3.2. How Agentic AI Affects Circumvention Pathways

AI agents are positioned to compress the most time-intensive steps in building synthetic supply chains. Filing incorporation documents across multiple jurisdictions currently requires human operators to navigate each jurisdiction's registry interface, prepare jurisdiction-specific documentation, and maintain consistency across filings. An agentic system with the multi-step task completion capabilities now demonstrated on benchmarks like GAIA (74.5 percent general task completion) and OSWorld (76.3 percent desktop task completion) could execute these workflows autonomously, generating internally consistent sets of trade documents, bills of lading, invoices, and end-user certificates, that vary enough to evade template-matching detection. Given the commercial availability of frontier models and the absence of procurement barriers on the circumvention side, adversaries face no adoption lag comparable to government procurement timelines of 18 to 36 months (Center for Strategic and International Studies, 2022).

Increased operational throughput does not hand circumvention a clean advantage. Circumvention networks can retry cheaply when an incorporation filing is rejected or a forged document fails to convince. These are operational failures, and they cost almost nothing. Detection failures are catastrophic. A single flagged shipment can trigger prosecution, asset seizure, and the unraveling of connected intermediaries. AI-assisted circumvention generates synthetic entities and documentation at higher volume, but each additional entity and transaction also creates a signal that enforcement could find. The same tools that let a network spin up twenty shell companies in a week also produce twenty new nodes for graph-based anomaly detection to flag. Whether circumvention gains ground depends on how fast enforcement can work through that expanded signal surface relative to the rate at which new synthetic fronts appear.

Physical transport remains irreducibly human but is not much of a barrier. Individual GPU modules are small, lightweight, and not subject to real-time serial number tracking. The suitcase method for Blackwell-series GPUs, disassembling server chassis and moving individual modules overland into southern China, shows that physical logistics are an inconvenience rather than a chokepoint. As AI sharpens KYC screening on the enforcement side, circumventors will respond with higher-quality synthetic documentation, and the contest will migrate back toward physical inspections and post-shipment verifications. Those are the stages where enforcement capacity is thinnest (Section 2).

3.3. Circumvention Bottlenecks

Despite the substantial capability gains that AI offers, contemporary circumvention remains constrained by a set of operational bottlenecks that are difficult to relax under present conditions. Understanding which of these bottlenecks AI can and cannot address is central to assessing the net impact on the enforcement-circumvention balance.

Trusted human intermediaries. Circumvention continues to depend on trusted human intermediaries at multiple stages of the supply chain. Procurement agents, warehouse operators, logistics coordinators, and financial handlers all occupy positions where malfeasance is difficult to automate and exposure carries asymmetric personal risk, including criminal prosecution and imprisonment. While AI systems can assist with screening, scheduling, and communication, they cannot substitute for the trust relationships required to physically handle controlled goods, falsify records at points of custody, or coordinate across jurisdictions without triggering suspicion. As networks expand to increase throughput, the number of participants with operational visibility grows, raising the probability of compromise through defection, error, or seizure of communications.

Physical handling requirements. Export-controlled chips must be stored, repackaged, shipped, and installed in real-world facilities. Each step introduces constraints related to transport capacity, inspection exposure, and logistical coordination. Even when shipments are disguised or misdeclared, they must traverse ports, customs facilities, and data centers subject to inspection and post-hoc verification. These requirements limit the extent to which circumvention can be virtualized and impose fixed costs that scale with volume.

Documentation coherence at scale. Documentation and identity fraud, while increasingly commoditized, introduces coordination constraints. Generating convincing end-user certificates, customs declarations, and corporate records at scale requires internal consistency across documents, entities, and transactions. As the number of shell companies and shipments grows, maintaining coherence becomes operationally complex. Errors or inconsistencies propagate across the network, increasing the likelihood that a single anomaly exposes broader patterns. AI-enabled throughput shifts the bottleneck from document quality to document coherence under scale, a constraint that graph-based anomaly detection on the enforcement side may be well-positioned to exploit.

Financial routing. Payments for high-value semiconductor transactions must ultimately move through regulated financial systems, often across multiple jurisdictions. Layering transactions to obscure end users increases operational complexity and latency, while concentrating payments raises detection risk. Operation Gatekeeper documented over $50 million in direct wire transfers from the PRC, routed through intermediate jurisdictions to obscure their origin (Department of Justice, 2025). Even with automated tooling, mismatches between declared destinations, counterparties, and payment origins remain a recurring point of failure.

Reconfiguration costs. While diversion networks can reorganize in response to new controls or enforcement actions, each cycle of reconstitution requires dissolving entities, rerouting logistics, and establishing new commercial histories. These reset cycles temporarily reduce throughput and degrade the trust relationships on which operations depend. The faster networks adapt, the more frequently they incur these costs, placing an upper bound on sustained diversion rates even under conditions of high automation.

3.4. Data Sovereignty as Defensive Capability

A critical and often overlooked defensive capability on the circumvention side is the legal framework within the PRC. China's data sovereignty regime, comprising the Data Security Law, Personal Information Protection Law, Counter-Espionage Law (as amended in 2023), and State Secrets Law (as amended in 2024), creates a legal sanctuary for diverted goods once they cross the border. These laws collectively prohibit providing data to foreign enforcement authorities without government approval and impose severe penalties for violations, ranging from administrative fines to criminal prosecution in extreme cases involving national security (National People's Congress, 2021). The practical effect is to prevent multinational corporations from complying with U.S. subpoenas or conducting internal investigations within their Chinese subsidiaries, forcing U.S. enforcement agencies to rely on external indicators and open-source intelligence, inherently less reliable than internal corporate data. This asymmetry in information access is a structural feature of the enforcement landscape that no AI capability can directly overcome.

3.5. The Bottleneck Persistence Framework

We now introduce the central analytical framework of this paper.

3.5.1. Classification Criteria

We classify a constraint as **AI-relaxable** if it satisfies all three of the following conditions: (1) *Information-processing character*: The constraint's binding force derives primarily from the cost, speed, or accuracy of processing, linking, or generating information. (2) *Demonstrated or near-term AI applicability*: Existing benchmarks, deployed systems, or expert forecasts provide evidence that AI systems can perform the relevant task at or near human-level quality within the 2026–2029 window. (3) *No non-informational dependency*: Relaxing the constraint does not require changes in legal authority, physical presence, jurisdictional reach, institutional coordination mechanisms, or procurement timelines.

A constraint is classified as **structurally binding** if relaxing it requires action outside the domain of information processing, i.e., if any of the three criteria above is not met. The key implication is that even unlimited computational capability cannot relax a structurally binding constraint.

3.5.2. Enforcement and Circumvention Compared

The bottlenecks AI relaxes on the circumvention side sit closer to the binding constraints of current smuggling operations than those it relaxes on the enforcement side. Circumvention today is constrained in large part by coordination across intermediaries, documentation coherence at scale, and multi-step planning capacity, all of which satisfy the AI-relaxability criteria above. Enforcement today is constrained in large part by physical verification capacity, legal authority, jurisdictional reach, and procurement timelines, none of which are information-processing problems. AI capability gains will therefore relieve a larger share of circumvention's binding constraints than enforcement's. Enforcement does benefit from AI, but the benefits concentrate in screening and triage, stages we term the "cheap layers" because AI can scale their throughput five- to tenfold. The "binding layers," converting flags into holds or denials, inspecting facilities, prosecuting violations, scale at roughly 1 to 1.5 times because they require legal authority, physical presence, and institutional coordination that computation cannot supply.

3.5.3. Estimating Bottleneck Composition

To summarize the framework's implications, we provide structured qualitative estimates of the share of binding constraints on each side that fall into AI-relaxable versus structurally binding categories. These estimates were produced through the consensus judgment process described in Section 1.5: each co-author independently assessed the relative weight of each bottleneck category based on enforcement-action records and practitioner interviews, and disagreements were resolved through discussion. We report ranges rather than point estimates to reflect the inherent uncertainty.

Enforcement side: approximately 35 percent of binding constraints are AI-relaxable (document processing, entity resolution, pattern detection, multilingual analysis); approximately 65 percent are structurally binding (physical verification, legal authority, jurisdictional reach, procurement timelines, detect-to-hold conversion).

Circumvention side: approximately 60 percent of binding constraints are AI-relaxable (shell company creation, document fabrication, multi-step coordination, counter-intelligence); approximately 40 percent are structurally binding (trusted human intermediaries, physical handling, financial routing through regulated systems).

These ranges are *structured qualitative judgments*, not empirically calibrated measurements. They should be read as indicating the direction and approximate magnitude of the asymmetry rather than as precise values. We specify disconfirmation criteria in Section 7.8.

4. Agentic Nearcast: How Capable AI Is Likely to Improve Offense-Defense Pathways

4.1. Advanced Agents as Enforcement/Circumvention Tools

Advanced autonomous AI agents represent a rapidly maturing class of AI capability with direct relevance to enforcement and circumvention workflows. In this section, we assess the current and near-term trajectory of specific agentic capabilities and evaluate their likely impact on the enforcement-circumvention balance through 2028. The scope of our analysis is restricted to agentic capabilities that satisfy the following selection criteria: *Direct workflow relevance*—we chose capabilities that map to specific tasks documented in BIS enforcement actions, DOJ prosecution, or think tank analyses of export enforcement. *Most transformative value*—we chose capabilities that we anticipate to change rapidly during the 2026–2028 window and remain a priority to frontier labs, inferred from benchmarks, expert forecasts, and broader deployment signals.

The impact consensus is measured using two confidence axes. First, we estimate confidence that agentic capability significantly evolves by 2028. Second, we measure confidence that the capability, once mature, can meaningfully impact the enforcement-circumvention balance. Our analysis is based on systematic review of frontier AI lab capability roadmaps and benchmark suites (METR, SWE-bench, WebArena) and a review of BIS enforcement actions and DOJ prosecutions to identify capability requirements for both enforcement and circumvention workflows. We constrain confidence range by the quality of available evidence: 75% and higher (Tier 1 evidence) corresponds to high-signal data from multiple expert/lab forecasts, benchmark trends, and deployed systems; 50–75% (Tier 2) corresponds to some expert views and benchmarks or direct deployment evidence; 50% and lower (Tier 3) primarily involves trend extrapolation, market signals, and speculative judgment.

4.2. Impact of Agentic Capabilities on Enforcement-Circumvention Balance

Table 3 summarizes the seven main capabilities assessed, their forecast confidence scores, and net directional impact. The three capabilities with the strongest structural asymmetries—Evidence Collection, Cybersecurity, and Autonomous Engineering & Planning—receive full analytical treatment in Sections 4.3.1–4.3.3.

Net Impact reflects the direction of tilt under default adoption trajectories. Though the split favors neither side decisively (four pro-enforcement, three pro-circumvention), the capabilities favoring circumvention target constraints closer to current operational bottlenecks.

4.3. In-Depth Discussion: Three Major Agentic Capabilities

4.3.1. Evidence Collection and Information Retrieval

Agentic open-source intelligence (OSINT) collection is already deployed at scale: Sayari processes 5.3 billion trade records across 100+ risk categories; Altana AI maintains a knowledge graph of 500+ million corporate entities with trade relationship mapping. By leveraging ML in processing check images and metadata continuously, the U.S. Treasury recovered $1 billion in check fraud in FY2024 alone. Graph Neural Networks (GNNs) for evidence collection achieve robust entity resolution even with limited labeled data.

We expect enforcement to greatly benefit from automated entity resolution, mainly by implementing agents cross-referencing corporate registries, names, addresses, and transliterations across languages and jurisdictions. Enforcement agencies also have the advantage of access to comprehensive government-protected databases: customs data, ownership records, and multilateral information channels. However, the same tools that help entity resolution can allow circumvention actors to conduct counter-intelligence, identifying which of their shell companies have been flagged, mapping enforcement agency investigative patterns, and monitoring overall vulnerability via OSINT platforms. This benefit is inherently defensive and reactive—circumvention actors can only detect it when they are already flagged, but cannot prevent the overall tightening of the enforcement net.

Net Impact: Favors *enforcement*. **Capability Forecast:** Growth: 85%, Significance: 60%.

4.3.2. Autonomous Engineering and Planning

AI agents can autonomously write, test, and deploy software and decompose complex objectives into sequential subtasks executed across digital environments. In mid-2025, frontier AI models resolved over 70% of real software engineering tasks on SWE-Bench Verified, up from approximately 20% in 2024 (Scale AI, 2026). METR benchmarks show that the time horizon of reliably completed tasks doubles approximately every seven months, accelerating to every four months in 2024–2025 (Kwa et al., 2025). Frontier models achieve comparable performance with classical planners on standard PDDL benchmarks (Corrêa et al., 2025). However, a complexity bottleneck persists: SWE-Bench Pro scores remain below 25% for enterprise-level tasks, and plan quality degrades substantially with horizon length.

Enforcement will benefit from agent-assisted pre-screening of end-use checks, automated case-management systems, and custom anomaly-detection pipelines. However, enforcement requires enterprise-grade reliability and auditability, bottlenecking deployment on technical implementation, procurement timelines, and inter-agency coordination. We expect circumvention actors to benefit from accelerated multi-step execution across company registration, document generation, transshipment logistics, and payment coordination, with fewer barriers to deployment: no procurement cycles, no auditability requirements, and readily available commercial infrastructure. Critically, circumvention tolerates higher error rates—if one shell company setup fails, another can be attempted at minimal loss.

Net Impact: Favors *circumvention*. **Capability Forecast:** Growth: 90%, Significance: 70%.

4.3.3. Cybersecurity and Adversarial Robustness

Agentic impact on cybersecurity remains actively debated. OWASP ranked prompt injection as the #1 critical LLM vulnerability despite the seemingly low intervention bar. Systematic red-teaming evaluations show attack success rates of 87.2% for GPT-4 and 82.5% for Claude 2, with jailbreak prompts transferring across model families at rates up to 64.1% (Pathade, 2025). A joint paper by researchers from AISI, Anthropic, and University of Oxford revealed that 12 published defenses against prompt injection were bypassed with attack success rates above 90% (Souly et al., 2025). Commercially, Deloitte projects AI-enabled fraud losses in the US to reach $40 billion by 2027.

Whether enforcement agencies deploy agents for screening or investigation, those agents will become targets for adversarial manipulation. Any enforcement AI that processes external documents (shipping records, corporate filings, trade data) will be vulnerable to adversarial manipulation, requiring human review at every decision point. We expect circumvention to deploy advanced agents to reveal and target vulnerabilities more effectively, jailbreaking models to generate fraudulent documentation, impersonation content, or operational planning assistance.

Net Impact: Favors *circumvention*. **Capability Forecast:** Growth: 90% (offense), 45% (defense). Significance: 75%.

4.4. Further Discussion—Capability Analysis

Although the offense-defense balance has no clear winner from the gains supplied by the advanced AI agents, the offense side holds a marginal edge. Offense's advantage is mostly driven by early adoption and larger error bandwidth. The analysis therefore flags the fundamental need for enforcement to (i) act upon inherent advantages and preemptively mitigate vulnerabilities identified by the Nearcast, and (ii) resolve systematic bottlenecks identified by the Nearcast (e.g., modernization of BIS infrastructure, early deployment of frontier technologies). The critical question is not which side gains more capabilities, but which side's *binding constraints* are more effectively relaxed.

Section 3.5 introduced the bottleneck persistence framework, which distinguishes constraints AI can relax (primarily informational: document processing, entity resolution, pattern detection, coordination) from those that remain structurally binding (physical verification capacity, legal authority, jurisdictional reach, procurement timelines). Mapping the seven capability assessments onto this framework reveals systematic asymmetries:

The binding constraints for circumvention are predominantly informational. Contemporary smuggling networks are constrained primarily by coordination across intermediaries, documentation coherence at scale, and multi-step planning capacity (Section 3.3). The capabilities that favor circumvention—Task Decomposition, Autonomous Multi-Step Execution, and Cybersecurity—directly target these informational bottlenecks.

The binding constraints for enforcement are predominantly structural. The capabilities that favor enforcement—Administrative Processing, Evidence Collection, Long-Horizon Monitoring, and Multilingual Comprehension—

improve screening, triage, and detection. These are genuine gains, but they operate in what this paper terms the “cheap layers” of enforcement. The binding layers remain structurally constrained regardless of computational capability.

This asymmetry is captured in the *detect-to-hold conversion gap*. AI-enhanced screening can generate flags at 5 to 10 times current throughput, but institutional mechanisms for converting flags into binding enforcement actions scale at only 1 to 1.5 times. The gap between detection capacity and action capacity is the structural feature that tilts the balance toward circumvention under default adoption trajectories.

4.5. Mapping Capabilities to Enforcement Workflows

To operationalize this framework for emerging capabilities beyond those assessed here, we classify the enforcement-circumvention contest into paired workflow functions. Table 5 presents these paired functions with their binding constraints.

4.6. Nearcast: Example Scenarios

We introduce example scenarios to showcase direct applicability of the Nearcast in real-world conditions for both sides of the offense-defense balance.

Example scenario, circumvention: creation of synthetic supply chains. By 2027 we predict that the manual creation of shell companies is likely to be largely automated through the use of capable AI agents that can produce vast volumes of high fidelity synthetic data, with diverse non-overlapping digital footprints. Key capabilities involved include automated incorporation via information retrieval agents navigating corporate registries of jurisdictions with lax oversight, production of synthetic documentation via generative AI models producing high-fidelity trade documents, and deepfake verification to bypass remote inspections or KYC checks.

Example scenario, enforcement: agent-assisted network interdiction from a single anomalous shipment. A first continuous monitoring agent operating on CBP trade data flags a pattern: three ostensibly unrelated resellers in Malaysia have each placed orders for small batches of H200 GPUs from different U.S. distributors within a 10-day window. Individually, each order falls below volume thresholds that would trigger enhanced screening. The monitoring agent cross-references corporate registries and identifies that all three resellers share a registered agent, were incorporated within six weeks of each other, and list directors whose names are transliteration variants of the same two individuals. Key capabilities involved include continuous monitoring identifying temporally clustered orders, cross-registry ownership mapping resolving transliteration variants, and automated escalation packet assembly for immediate OEE investigator review.

4.7. Conclusion

The model scenarios above illustrate the two faces of the agentic transformation. In the first, AI relaxes circumvention’s informational bottlenecks (coordination, documentation, identity fabrication) directly increasing throughput against constraints that are close to current operational limits. In the second, AI delivers genuine enforcement gains in detection and triage, but the binding constraints that determine outcomes (investigator capacity, hold authority, reconstitution speed) remain structurally unchanged. The gap between what AI enables and what institutions can act on is the central dynamic that shapes the balance.

5. 2027–2029 Dynamics and Enforcement-Circumvention Balance Outcomes

Section 4 mapped which agentic capabilities plausibly mature by 2027 to 2029 and how they change specific enforcement and circumvention tasks. This section translates those capability shifts into (i) a concrete repeated-game timeline of checks and counter-checks, and (ii) an assessment of which side gains a durable advantage under different adoption postures.

The central insight is that detection matters only when it quickly leads to a shipment or access being paused. Document checks can take hours or a few days. A pause or denial can take days. On-site visits often take weeks. Penalties often take months. When enforcement cannot turn flags into real action quickly enough, AI agents increase screening faster than enforcement can actually stop suspect shipments or access. The result is a system where more transactions look compliant on paper, without a matching reduction in diversion.

5.1. Sequence of Likely Moves and Counter-Moves Under High AI Agent Adoption (2027–2029)

Table 6 is illustrative rather than predictive. It shows plausible recurring patterns of adaptation under high agent adoption, not a claim about exact sequencing or a single common pathway.

Table 6 shows a recurring pattern rather than a fixed sequence: AI is likely to improve paperwork review, screening, and narrative checking faster than it improves the harder parts of enforcement, such as pausing transactions, conducting inspections, and sustaining post-shipment control. Section 5.2 therefore turns from interaction patterns to scenario analysis, asking under what adoption conditions these improvements change outcomes rather than simply increasing screening volume.

5.2. Assessing the Enforcement–Circumvention Balance

5.2.1. Main Finding

Under current conditions, more capable AI agents are likely to tilt the balance toward circumvention because they make it easier to create transactions that appear compliant, while increasing screening faster than enforcement can turn flags into real action. The balance moves toward enforcement only when three conditions hold simultaneously. First, checks must be tied to the points where a transaction can still be stopped (payment release, shipment booking, or compute access) so that failure means the transaction cannot proceed. Second, cases that fail key checks should rarely be allowed to proceed, and when they are, those exceptions should be logged, reviewed, and carry real consequences. Third, high-confidence flags must reliably lead to holds, denials, inspections, or penalties within days to weeks. The main uncertainty is whether these improvements actually reduce diversion, or mainly push it into post-shipment channels where visibility and cross-border leverage are weaker.

5.2.2. Methodology

We use three steps to move from capability analysis to scenario assessment. **Step 1: Map capabilities onto enforcement controls.** We identify where AI agents affect specific controls in the enforcement chain and ask whether those effects improve screening only or also improve the ability to pause, deny, inspect, or penalize suspicious transactions. **Step 2: Assess how feasible adoption is.** We evaluate how likely different kinds of enforcement adoption are using a tiered adoption model and a six-factor feasibility rubric. **Step 3: Build scenarios and update conditions.** We combine these judgments into four scenarios, each representing a different adoption posture across BIS, exporters, and enforcement partners.

5.2.3. Net Advantage Score

The net advantage score is a rough overall assessment of whether conditions in a given scenario favor enforcement or circumvention. It ranges from -2 (circumvention-leaning) to $+2$ (enforcement-leaning), with 0 representing approximate equilibrium. The score is anchored to three observables: diversion success rate, detect-to-hold conversion speed (within days to weeks, pre-shipment or pre-access), and displacement share (pre-shipment diversion shifting to post-shipment channels).

5.2.4. Scenarios A–D: Adoption Patterns and Likely Outcomes

In Table 7, each scenario varies along two main questions: first, whether checks are tied to the points where a transaction can still be stopped, and second, whether enforcement can follow through at scale once suspicious cases are flagged. A common assumption across all four scenarios is that circumvention actors adopt advanced agentic capabilities at high levels across 2027–2029.

5.2.5. Assumptions and Indicators to Watch

The scenarios above rest on a small number of assumptions that can be tracked over time. Key assumptions include: (1) checks are tied to key stopping points, (2) exceptions are rare, logged, and costly, (3) follow-through scales with targeting, and (4) later transfers become more visible and more enforceable. What would change the assessment: sustained increases in blocked controlled volume before shipment or access, without a corresponding rise in post-shipment leakage.

5.2.6. Why Post-Shipment Is Structurally Hard

Post-shipment diversion is hard to control because enforcement becomes weaker once goods move across borders and into secondary markets. Legal authority narrows, visibility falls, and the number of intermediaries grows. At the same time, the evidence becomes noisier: legitimate repairs, resale, leasing, and relocation can all make suspicious movements look routine, while warranty, return and resale signals often arrive late or incompletely.

5.2.7. What Could Shift the Balance

The analysis points to four changes that matter most. First, checks need to be tied to the points where a transaction can still be stopped, such as payment release, shipment booking, or access control. Second, failed checks should rarely be waived, and when exceptions are made, they should be logged, reviewed, and carry real consequences. Third, better targeting needs to be matched by faster follow-through. Fourth, later resale, leasing, and transfer need to become more visible and more enforceable.

6. Implications for Enforcement Architecture

The analysis points to a conclusion: better AI tools alone will not meaningfully improve export control enforcement unless the surrounding system changes as well. AI can help agencies and firms review more documents, flag more suspicious cases, and connect more signals across records. But those gains matter only if the system can also stop risky transactions in time, verify what is happening on the ground, retain visibility after shipment, and coordinate across the many actors involved in enforcement.

6.1. Turning Flags into Action

The most important bottleneck identified in this paper is the gap between spotting a suspicious case and actually stopping it, reflected by the detect-to-hold conversion gap. Checks need to be tied to the points where a transaction can still be stopped: payment release, shipment booking, or compute access. If key facts cannot be verified at those points, the transaction should not move forward. Instead of asking enforcement to catch every bad case after the fact, the system places more of the burden on the seller, intermediary, or access provider to verify basic facts before the transaction can proceed.

6.2. Physical Verification Capacity

BIS currently operates with approximately 9 to 11 ECOs overseas and approximately 150 domestic investigators, conducting on the order of 100 PLCs per year (Practitioner interviews, 2025–2026). This capacity is structurally insufficient to verify the volume of controlled chip transactions occurring through transshipment hub jurisdictions. AI cannot substitute for physical presence at a facility. One practical benefit of AI is that, if it automates more of the document-heavy screening, triage, and case-preparation work now done by analysts and officers, BIS can redirect more staff time toward the physical tasks that AI cannot perform well, such as site visits, end-use checks, and follow-up on suspicious facilities. Another measure is deeper cooperation with allied governments in places that matter most for transshipment. Hardware-backed attestation mechanisms, including cryptographic device identity and GPU location verification (Nellis & Martina, 2025; Fist et al., 2024), represent a complementary approach.

6.3. Post-Shipment Observability

Post-shipment diversion is structurally hard because jurisdiction and authority break after cross-border transfer (Section 5.2.6). Our scenario analysis shows that even scenarios where pre-shipment interdiction improves can produce net-zero diversion reduction if displacement into post-shipment channels is not addressed. If controlled items are subject to mandatory reporting at each transfer of ownership or location, the information asymmetry that currently shields post-shipment diversion would narrow.

6.4. Procurement Timeline Compression

Government AI procurement timelines of 18 to 36 months (Center for Strategic and International Studies, 2022) represent a structural disadvantage in a contest where adversaries can adopt commercially available tools immediately. Addressing this bottleneck requires either compressing procurement timelines through expedited authorization pathways for enforcement AI tools, or shifting enforcement to rely more heavily on private-sector compliance systems that can adopt AI more quickly.

6.5. Better Coordination Across Agencies and Firms

Enforcement is only as strong as the coordination between its main actors. Export enforcement is spread across licensing, customs, investigations, prosecutors, exporters, distributors, cloud providers, and, in some cases, foreign partners. Without this coordination layer, AI is likely to produce better local optimization inside separate organizations without creating a stronger overall enforcement system.

7. Limitations

7.1. Uncertainty

Several important questions remain unresolved. No official U.S. government estimate exists for what percentage of export control violations are detected, though practitioners generally believe the figure to be low given the volume of transactions relative to enforcement capacity (Practitioner interviews, 2025–2026). No comprehensive assessment measures whether prosecutions reduce smuggling attempts (the deterrence question). The timeline for BIS deployment of modern AI capabilities remains unclear. The trajectory of Chinese state involvement is uncertain.

7.2. Adoption Heterogeneity

The scenarios in Section 5 assume that circumvention actors adopt agentic capabilities at uniformly high levels. In practice, adoption is likely to be uneven across networks. The assessment is therefore best read as describing the upper bound of the circumvention advantage under conditions of widespread adoption.

7.3. Model Access as a Variable

If frontier model providers restrict access for circumvention-related tasks, this changes the capability landscape. However, smaller open-weight models with fewer restrictions are widely available, and fine-tuning can circumvent many safety measures. We treat model access restrictions as a friction that slows but does not prevent circumvention-side adoption.

7.4. Chinese Domestic Production

This analysis assumes that controlled chips remain primarily available through U.S. and allied supply chains. If Chinese domestic chip production reaches near-frontier capability levels, the enforcement–circumvention contest changes character fundamentally. We treat this as a boundary condition rather than a near-term likelihood.

7.5. Risk of Agentic Deployment

Deploying agentic systems for enforcement introduces new attack surfaces. Graph poisoning, in which adversaries inject fabricated entities or relationships into trade-network databases to deceive graph-based detection systems, becomes more feasible as generative AI reduces the cost of producing convincing forgeries.

7.6. Methodological Limitations

This analysis is informed by practitioner interviews, public enforcement records, benchmark data, and the analytical framework developed herein, rather than by classified intelligence or comprehensive empirical data on circumvention volumes. The net advantage scores in Section 5 are qualitative indices reflecting the authors' best judgment, not empirically calibrated measurements.

7.7. Disconfirmation Criteria

Our central conclusion, that the default trajectory favors circumvention, would be disconfirmed by evidence of sustained increases in blocked controlled volume pre-shipment without a corresponding rise in post-shipment transfer leakage. If this pattern were observed over a 12 to 18 month period, it would indicate that AI-enhanced enforcement is producing net reductions in diversion rather than merely displacing it, and our assessment would require revision.

8. Conclusion

Advanced AI agents will improve enforcement efficiency in targeted domains: administrative processing, anomaly detection, multilingual analysis, and continuous monitoring. But these gains are concentrated in the cheap layers of enforcement, screening and triage. The binding layers that determine outcomes, physical verification, legal authority to convert detection into interdiction, and post-shipment observability, remain structurally constrained and largely resistant to computational solutions.

On the circumvention side, AI relaxes constraints that are closer to the binding limits of current operations: coordination, documentation, and planning capacity. The result, under default adoption trajectories, is that the enforcement-circumvention balance shifts further toward offense. AI makes the contest faster and more complex without changing its fundamental structure. This dynamic is analogous to patterns documented in cybersecurity, where each generation of defensive technology is met by offensive adaptation, and the structural conditions of the contest, not the capabilities of any single tool, determine the long-run balance (Garfinkel & Dafoe, 2019; Slayton, 2017).

The path forward requires treating AI as a force multiplier for fundamentally restructured enforcement architecture, not as a solution in itself. Hard-stop checks coupled to chokepoints, expanded physical verification through allied cooperation, transfer-event reporting to prevent post-shipment displacement, and compressed procurement timelines constitute the structural interventions necessary to shift the underlying equilibrium. Without these parallel reforms, agentic AI will accelerate enforcement burn rate, the exhaustion of investigative resources chasing multiplying false fronts, without proportionally increasing interdiction capacity. Only by altering the structural conditions of the contest can policy restore a defensible equilibrium in export control enforcement.

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Table 1. Key terms and definitions. Terms marked with † are introduced in this paper; remaining terms adapt existing usage to the export control context.

Term	Definition
Agentic AI	AI systems capable of autonomously planning and executing multi-step actions in pursuit of a goal without continuous human oversight, distinguished from conventional AI tools by their capacity to decompose objectives into subtasks, execute them sequentially across digital environments, and adapt based on intermediate results.
Functional risk	The possibility that AI systems act as autonomous operatives executing cyber, logistical, or procurement operations with minimal human direction, rather than serving as productivity tools for human operators. Treated here as a plausible near-term development, not an established fact (Withers, 2025; Rasser, 2026).
Offense–defense balance	The degree to which structural conditions systematically favor circumvention (offense) or enforcement (defense) in a regulatory contest. Adapted from international security theory (Jervis, 1978; Garfinkel & Dafoe, 2019) to the export control context.
AI-relaxable constraint	A constraint satisfying three conditions: (i) its binding force derives primarily from the cost, speed, or accuracy of information processing; (ii) existing or near-term AI systems demonstrate applicable capability; and (iii) its relaxation requires no changes in legal authority, physical presence, jurisdictional reach, or institutional coordination.
Structurally binding constraint	A constraint that persists regardless of computational capability because its relaxation requires action outside the domain of information processing. Examples include physical verification capacity, legal authority, jurisdictional reach, and government procurement timelines.
Bottleneck composition	The estimated share of each side's binding constraints falling into AI-relaxable versus structurally binding categories. Enforcement: approximately 30–40% AI-relaxable, 60–70% structurally binding. Circumvention: approximately 55–65% AI-relaxable, 35–45% structurally binding. These are structured qualitative judgments, not empirically calibrated measurements (Section 3.5.3).
Cheap layers / Binding layers†	A paired distinction within the enforcement pipeline. *Cheap layers* (screening, anomaly detection, flagging) are stages where AI increases throughput substantially ($\sim$5–10$\times$) because they are primarily informational tasks. *Binding layers* (holds, denials, physical inspection, penalties, post-shipment verification) are stages where throughput scales slowly ($\sim$1–1.5$\times$) because they are constrained by legal authority, physical presence, and institutional coordination.
Binding actions	Enforcement outputs that materially constrain diversion: shipment holds, license denials, seizures, civil or criminal penalties, Entity List designations, and revocation of export privileges. Distinguished from screening and detection, which identify suspicious activity but do not independently prevent it.[1]
Detect-to-hold conversion gap†	The structural disparity between the rate at which AI-enhanced screening generates flags and the rate at which institutional mechanisms convert those flags into binding actions.
Agentic nearcast†	A short-horizon analytical exercise adapted from forecasting terminology (Karnofsky, 2022) that maps emerging agentic capabilities onto specific workflows to assess their near-term (2026–2029) differential impact.
Displacement share	The proportion of diversion activity shifting from pre-shipment channels (where enforcement is stronger) to post-shipment channels (resale, leasing, transfer of operational control) where jurisdiction and observability are structurally weaker.

Table 2. U.S. export control enforcement architecture for controlled AI chips, illustrating the progression from licensing to post-shipment verification.

Step	Action	Agency	Detail	Timeframe
Stage 1: Licensing & Authorization				
1	License application (SNAP-R)	BIS	Classification, destination, buyer, end-use	Days to weeks
2	Initial review	BIS (Commerce)	Completeness, accuracy, risk indicators	
3	Interagency review	DoD, DoE, State	Security, foreign policy, proliferation risk	
4	Pre-license checks (optional)	BIS Field Office	On-site verification of buyer legitimacy	
5	License decision	BIS	Approved, denied, or returned without action	
Stage 2: Export Execution				
6	File EEI in AES	Exporter	Exporter identity, buyer, ECCN, license number	Minutes to hours
7	Outbound targeting & inspection	CBP	Automated screening; physical inspection if flagged	
8	Shipment leaves U.S.	—	Goods exported to declared destination	
Stage 3: Post-Shipment Enforcement				
9	Post-shipment verification	BIS, U.S. Embassy	Site visits confirming item present and as declared	Weeks to years
10	Investigation of violations	BIS/OEE	Diversion, smuggling, false documentation	
11	Enforcement outcomes	DOJ, BIS	Civil penalties, prosecution, entity listing	

Table 3. Agentic capability assessment. Growth measures confidence that the capability significantly evolves by 2028. Impact measures confidence that the mature capability meaningfully shifts the enforcement-circumvention balance. Net Impact reflects the direction of tilt under default adoption trajectories.

Capability	Growth	Significance	Net Impact	Rationale
Administrative Processing & Record-Keeping	5	3	Enforcement	Network-level document analysis accessible only to entities with comprehensive trade data.
Task Decomposition	4	3	Circumvention	Faster adoption, higher error tolerance, more manual workflows to automate.
Evidence Collection & Information Retrieval	5	4	Enforcement	Government access to protected databases creates structural advantage.
Autonomous Engineering & Planning	5	4	Circumvention	No procurement barriers, tolerates failure, plans against slow, observable opponent.
Cybersecurity & Adversarial Robustness	5 (off.) 3 (def.)	5	Circumvention	Fundamental attacker-defender asymmetry: offense needs one breach, defense $\sim$100%.
Long-Horizon Monitoring	4	3	Enforcement	Requires persistent access to comprehensive data streams only government possesses.
Multilingual Comprehension	5	3	Enforcement	Addresses enforcement's language-barrier constraint; circumvention already bilingual.

Table 4. Confidence scoring rubric.

Score	Tier	Evidence Criteria
5	Decisive	Multiple independent lines converge.
4	Strong	At least two independent evidence lines.
3	Moderate	One solid evidence line with reasoning.
2	Weak	Indirect or analogical evidence only.
1	Minimal	Speculative; no direct evidence.

Table 5. Defense–offense workflow pairs and binding constraints.

Defense Workflow	Offense Counterpart	Binding Constraint
Verification: validating identity, end-use, and documentation claims.	**Fabrication**: creation and curation of fake entities, records, and traces.	Physical verification capacity; fabrication quality detectable only at scale through graph-based analysis.
Monitoring: continuous tracking and anomaly detection over time.	**Signal suppression**: degrading observability through telemetry poisoning and noisy signals.	Data infrastructure modernization; monitoring is only effective when coupled to binding follow-through.
Investigation: tracking active cases and building profiles.	**Deception**: evidence destruction or engineering to mislead analysis.	Legal authority and inter-agency coordination; investigation timelines measured in months vs. circumvention cycles in days.
Pattern Detection: analyzing signals, linking records.	**Segmentation**: promoting record segmentation to prevent entity resolution.	Data integration across jurisdictions; segmentation exploits fragmentation AI alone cannot resolve.
Interdiction: applying enforcement task chains to block attacks.	**Capture**: shifting defender's decisions by compromising system controls.	Legal authority and jurisdictional reach represent the ultimate binding constraint.
Guarding: cyberdefense and system integrity.	**Disruption**: active system attacks to disable cyberdefenses.	Fundamental attacker-defender asymmetry; defense costs scale with attack surface.

Table 6. Structural interaction dynamics under high AI agent adoption (2027–2029). Each dynamic is grounded in analogous patterns from sanctions evasion, cybersecurity, and regulatory enforcement literatures.

Dynamic	Mechanism	Historical Analogue	Testable Implication
Screening–compliance arms race	AI-enhanced screening raises the baseline for "plausible compliance," compelling higher-fidelity synthetic documentation.	Russia sanctions evasion: entity proliferation outpacing investigative capacity (Bilousova et al., 2024).	Rising per-transaction screening quality coexisting with stable or rising aggregate diversion volumes.
Detection–action decoupling	AI scales detection throughput (5–10×) faster than institutional mechanisms convert flags into binding actions (∼1–1.5×).	"Alert fatigue" in AML: SAR volumes grew post-9/11 without proportional follow-through.	Flag volumes rising while hold-to-flag ratios decline, average time from flag to action increasing.
Verification displacement	As pre-shipment controls tighten, diversion shifts toward post-shipment pathways where jurisdiction and observability are weaker.	"Balloon effect" in narcotics interdiction (Friesendorf, 2007); sanctions enforcement displacement (Early, 2015).	Declining pre-shipment diversion coexisting with rising secondary-market transfers.
Adversary reconstitution advantage	Networks dissolve and reconstitute entities faster than enforcement can investigate and designate.	"Whack-a-mole" dynamic in counter-proliferation; cyber-threat infrastructure reconstitution (Gartzke & Lindsay, 2015).	Shrinking average lifespan of flagged entities combined with stable aggregate network throughput.

Table 7. Scenario analysis: adoption patterns and likely outcomes (2027–2029).

	Adoption Posture	Flag→Action	Score	Probability	Why
A	BIS: low–med; exporters: low–med	Little coordination at stopping points	-1 (to -2)	Likely	Circumvention improves compliance appearance. Enforcement screens better but cannot stop transactions quickly.
B	BIS: high; exporters: low–med	Government tools improve but checks weakly tied to stopping points	-1 (to 0)	Plausible	Better targeting creates more flags, but exporters do not stop transactions consistently. More leads without matching action.
C	BIS: low–med; exporters: high	Major channels strengthen checks and escalation; gov. follow-through limited	0 (to $+1$)	Unlikely / Plausible	Large channels block more risky transactions pre-shipment, but circumvention reroutes to weaker channels.
D	BIS: high; exporters: high	Checks tied to real stopping points; follow-through improves at scale	$+1$ (to 0)	Unlikely	Only posture that reliably turns flags into holds, denials, inspections, or penalties. Gains remain corridor-specific.

A. Cloud and Service-Based Circumvention

This appendix examines a complementary circumvention vector, the provision of advanced AI compute through cloud and service-based delivery models, which illustrates how delivery-model shifts compound the structural advantages identified in our main analysis. When physical export of restricted AI chips is constrained, access to compute can be obtained remotely by leasing time on foreign-located servers equipped with controlled hardware, decoupling physical possession from effective use.

A.1. Characteristics of Cloud-Based Access

Export control regimes have historically treated the transfer of physical items as the central object of regulation. Cloud-based compute complicates this regulatory framework by allowing the functional benefits of controlled hardware to be accessed without a transfer of possession. Under earlier regulatory interpretations, the provision of compute time or remote access to hardware was not unambiguously classified as an export of the underlying technology. Once compute access is abstracted into a service, physical chokepoints such as customs inspections and shipment monitoring no longer apply, and enforcement shifts toward contract oversight, user attribution, and network-level monitoring.

A.2. Service-Based Circumvention Modalities

Cloud-based circumvention occurs through several distinct service layers. Infrastructure-as-a-Service and Hardware-as-a-Service platforms allow users to rent dedicated GPU capacity or bare-metal access to accelerators hosted in foreign data centers. Higher-level services further abstract the underlying hardware. Software-as-a-Service AI tools perform computation on provider-managed servers, while API-based access to large models allows users to benefit from training and inference conducted entirely on controlled hardware. In these cases, the end user may never directly interact with GPUs, complicating attribution.

A.3. Regulatory Response

U.S. policymakers have moved to address these pathways. The Remote Access Security Act, passed by the House in January 2026, extends export control coverage to the provision of controlled compute resources to restricted jurisdictions. Complementary executive actions emphasize enhanced end-use monitoring and location verification. Industry responses have included software-based GPU location verification mechanisms. These measures collectively aim to align legal definitions with the operational realities of networked compute delivery. However, closing the legal gap does not resolve the enforcement challenge: detecting and proving violations in cloud environments requires new workflows that rely heavily on private-sector compliance and indirect indicators of misuse.

A.4. Residual Evasion and Structural Limits

Even under expanded legal authority and improved provider compliance, residual circumvention pathways are likely to persist. Adversaries can employ intermediaries in neutral jurisdictions, compromise legitimate cloud accounts, or distribute workloads across many lower-tier instances to avoid detection thresholds. Emerging decentralized compute marketplaces, which match demand and supply for GPU cycles on a peer-to-peer basis, further complicate enforcement.

A.5. Assessment

Measures targeting cloud-based circumvention raise the cost and complexity of accessing restricted AI compute but do not eliminate access. Enforcement effectiveness depends on sustained provider cooperation, allied alignment, and the ability to attribute service usage to ultimate beneficiaries. As delivery shifts further toward services and abstraction layers, the offense retains structural advantages in speed, adaptability, and deniability, while enforcement remains constrained by jurisdictional boundaries and limited observability. This pattern, enforcement gains that raise costs without closing pathways, mirrors the broader dynamic identified in our main analysis and reinforces the conclusion that structural measures, not screening improvements alone, are necessary to shift the equilibrium.